

SEMESTER END EXAMINATIONS - APRIL / MAY 2023

Program	: B.E :- Computer Science and Engineering	Semester	: III
Course Name	: Discrete Mathematical Structures	Max. Marks	: 100
Course Code	: CS35	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Explain substitution rule with an example Use the logical equivalences to show that $\neg (p \vee \neg (p \wedge q))$ is a contradiction. CO1 (06)
 - Show that 'b' can be derived from the premises $a \rightarrow b$, $c \rightarrow b$, $d \rightarrow (a \wedge c)$, d by method of contradiction. CO1 (06)
 - Define open statement with an example. With the universe of entire student body in a particular college and with the following open statements:
J(x): x is a junior, s(x): x is a senior, p(x): x is enrolled in a physical education class, represent each of the following arguments in the symbolic form and prove the validity of the argument.
No junior or senior is enrolled in a physical education class
Mary Gusberti is enrolled in a physical education class.
Therefore, Mary Gusberti not a senior. CO1 (08)
- Solve the following logical equivalence using laws of logic: CO1 (06)
i. $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \Leftrightarrow \neg (q \vee p)$ ii. $[(-p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \Leftrightarrow p \wedge q$.
 - Prove the validity of the following arguments: CO1 (09)
(i) $(\neg p \vee q) \rightarrow r, r \rightarrow (s \vee t), \sim s \wedge \sim u, \sim u \rightarrow \sim t, \therefore p$
(ii) $[(p \rightarrow q) \wedge (\sim r \vee s) \wedge (p \vee r)] \rightarrow (\sim q \rightarrow s)$
(iii) $p \rightarrow (q \rightarrow r), p \vee s, t \rightarrow q, \sim s, \therefore \sim r \rightarrow \sim t$
 - Let p(x), q(x) and r(x) be the following open statement: CO1 (05)
p(x): $x^2 - 7x + 10 = 0$
q(x): $x^2 - 2x - 3 = 0$
r(x): $x < 0$
Determine the truth value of the following statements if the universe is of all integers. Provide a counter example for the false statement.
(i) $\forall x [p(x) \rightarrow \sim r(x)]$
(ii) $\forall x [q(x) \rightarrow r(x)]$
(iii) $\exists x [q(x) \rightarrow r(x)]$
(iv) $\exists x [p(x) \rightarrow r(x)]$

UNIT - II

- Show that the divides relation on the set of integers is not an equivalence relation. CO2 (06)
 - Define partitions of a set with an example. $A = \{(-4, -20), (-3, -9), (-2, -4), (-1, -3), (1, 2), (1, 5), (2, 10), (3, 6), (4, 8), (4, 12)\}$ Define the relation R on A. by (a,b) R (c,d) if $ad=bc$.
Verify that R is an equivalence relation on A. find the equivalence classes: $[(-3, -9)]$, $[(4, 8)]$ and $[(1, 5)]$. CO2 (08)
 - With an example define the following: CO2 (06)
i. Composition of relation ii. Antisymmetric iii. partition of a set.

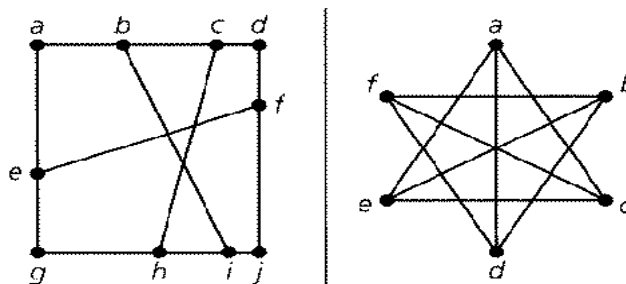
4. a) Define total order relation .Let $A = \{1, 2, 3, \dots, 12\}$ and let $D = \{(x, y) \in A \times A : x \mid y\}$. Then D is a partial ordering of A . Draw Hasse diagram for the poset (A, D) . CO2 (06)
- b) For the poset $(\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}, \mid)$, draw the hasse digram find the following: CO2 (10)
- Maximal element(s)
 - Minimal element(s)
 - Greatest element, if it exists
 - Least element, if it exists
 - Upper bound(s) of $\{2, 9\}$
 - Least upper bound of $\{2, 9\}$, if it exists
 - Lower bound(s) of $\{60, 72\}$
 - Greatest lower bound of $\{60, 72\}$, if it exists.
- c) Under what conditions is $A - B = B - A$? CO2 (04)

UNIT - III

5. a) In how many ways can one arrange all of the letters in the word INFORMATION so that no pair of consecutive letters occurs more than once?[Here we want to count arrangements such as IINNOOFRMTA and FORTMAIINON but not INFORINMOTA(where "IN" occurs twice) or NORTFNOIAMI (where "NO" occurs twice).] CO3 (08)
- b) Find the rook polynomial for the standard 3×3 chess board using expansion formula. CO3 (06)
- c) Find the number of arrangements of the letters in TALLAHASSEE. How many of these arrangements have no adjacent A'S. CO3 (06)
6. a) A total amount of Rs.1500 is to be distributed to 3 poor students A,B,C of class .In how many ways the distribution can be made in multiples of Rs.100. CO3 (08)
- if everyone of these must get at least Rs. 300.
 - if A must get atleast Rs. 500 and B & C must get atleast Rs 400 each.
- b) Define the derangement? Find the number of derangements of 1, 2, 3, 4.List all the derangements. CO3 (06)
- c) Find the number of permutations of a,b,c,dx,y,z, in which none of patterns spin, game, path or net occurs. CO3 (06)

UNIT- IV

7. a) Explain the following with an example: CO4 (06)
- Homeomorphic graphs
 - Hamilton cycle
 - chromatic number.
- b) Determine which of the following graphs are planar .If graph is planar; redraw it with no edges overlapping. If it is nonplannar, find a sub graph homeomorphic to either K_5 or $K_{3,3}$. CO4 (06)

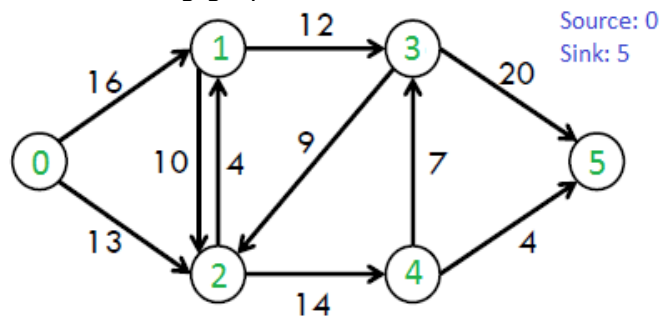


- c) Prove that in graph there is u-v trail if and only if there is a u-v path. CO4 (08)

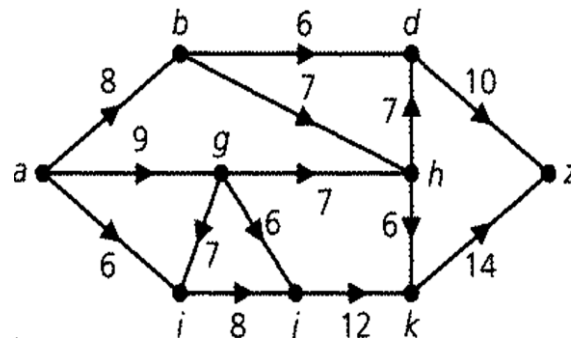
8. a) Define the following with an example: CO4 (06)
 (i) Complete Bipartite (ii) induced sub- graph (iii) Isomorphic graphs.
 b) Define Euler circuit with an example. Construct an Euler circuit for the following graph. CO4 (06)
 c) Define chromatic number of graph .Find the chromatic polynomial for C5 (A graph with cycle of length =5). CO4 (08)

UNIT - V

9. a) Define Rooted tree with example. Which of these codes are prefix codes? CO5 (04)
 i.a: 11, e: 00, t: 10, s: 01
 ii.a: 0, e: 1, t: 01, s: 001
 iii.a: 101, e: 11, t: 001, s: 011, n: 010
 iv.a: 010, e: 11, t: 011, s: 1011, n: 1001, i: 10101
 b) Explain Ford-Fulkerson Algorithm for Maximum Flow Problem. Find the maximum in the following graph from node 0 to node 5. CO4 (08)



- c) Obtain an optimal prefix code for the message MISSION SUCCESSFUL. CO5 (08)
10. a) Prove that a tree with n vertices had $n-1$ vertices. CO5 (04)
 b) Find the maximum flow and the corresponding minimum cut in the network below. CO4 (08)



- c) Define optimal prefix code. Use Huffman coding to encode these symbols with given frequencies: A: 0.10, B: 0.25, C: 0.05, D: 0.15, E: 0.30, F: 0.07, G: 0.08. What is the average number of bits required to encode a symbol? CO5 (08)
